

QUANTIFYING THE ANTHROPOGENIC AND CLIMATIC IMPACTS ON VEGETATION DYNAMICS IN SAMARKAND PROVINCE, UZBEKISTAN

Final Project Report

A climate-controlled attribution using MODIS NDVI, MODIS LST, CHIRPS precipitation, ERA5-Land solar radiation and the Global Human Footprint Index, with Random Forest cross-validation (2015–2024)

Samarkand Province, Uzbekistan (2015–2024)

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Abstract

Vegetation in semi-arid Central Asia responds to both climate variability and human land use, and the two are difficult to separate from observation alone. This study quantifies their relative contributions to vegetation change in Samarkand Province, Uzbekistan, over 2015–2024. MODIS MOD13Q1 Collection 6.1 NDVI (vegetation response), MODIS MOD11A2 Collection 6.1 land surface temperature, CHIRPS precipitation and ERA5-Land solar radiation (climate drivers), and the annual Human Footprint Index (HFI; anthropogenic driver) were harmonized to a 1 km grid, restricted to vegetated land and aggregated to growing-season (April–October) annual values for 20,132 pixels. Per-pixel trends were estimated with Theil–Sen slopes and Mann–Kendall tests. A RESTREND regression isolated the non-climatic NDVI signal, a partial-derivative decomposition split the NDVI trend into climate (CC) and human (HA) contributions to produce a six-class attribution map, and the result was cross-checked with a Random Forest model and a residual-versus-HFI comparison. The HFI layers for 2019–2024 come from later releases of the Mu et al. dataset that are not described in the peer-reviewed paper and contain a marked 2018→2019 step, so all HFI-dependent results were recomputed under alternative treatments of that break.

Climate, led by land surface temperature, was the dominant control on NDVI: the per-pixel median climate share of the attributable trend was 82% (HA 18%), and the Random Forest assigned the climate variables 90.4% of the importance among the physical drivers. Both figures were stable in the sensitivity analysis (median HA share 18–19%; HFI importance 7.7–9.6%). Zonal statistics of the rasters show a clear west–east gradient. The grassland-dominated western lowlands (west of ~66.1°E) show NDVI decline (median Sen slope -0.0034 to -0.0048 NDVI yr^{-1}), negative residual trends and poor climate-model fit (median R^2 0.16–0.43), and contain 64% of the 4,030 pixels classed as human-dominated degradation (20.0% of valid pixels). The cropland-dominated central and eastern districts show greening (median slope up to $+0.0032$ NDVI yr^{-1}) and contain 96% of the climate-dominated restoration pixels (26.2%). The residual–HFI correlation was positive (Spearman $\rho = +0.383$) but fell to $+0.17$ once the 2018→2019 HFI step was removed, and the western degradation label rests on a weakly constrained HFI coefficient. The western decline is therefore robustly non-climatic, but its attribution specifically to human pressure, and any link between HFI growth and irrigation-driven greening, remain provisional.

Keywords: vegetation dynamics; NDVI; RESTREND; climate–human attribution; Human Footprint Index; Random Forest; remote sensing; Samarkand; Central Asia.

1. Introduction

1.1 Motivation

Vegetation cover is one of the most sensitive indicators of land-surface change, integrating the combined influence of climate variability and human pressure into a single observable signal. In semi-arid regions, this sensitivity is particularly acute: small shifts in temperature, precipitation, or anthropogenic activity can tip ecosystems between greening and degradation trajectories with consequences for soil stability, water resources, agricultural productivity, and rural livelihoods. Central Asia exemplifies this vulnerability. The region has warmed roughly twice as fast as the global average over the past three decades, its hydrological regime is dominated by snow-fed rivers under increasing pressure from upstream withdrawals, and its dryland and oasis ecosystems coexist with rapidly expanding human settlements and intensifying land use.

Samarkand Province in Uzbekistan offers a particularly informative setting for studying these dynamics. The province lies at the climatic and ecological transition between irrigated agriculture along the Zarafshan River and the surrounding semi-arid rangelands and foothill steppe. Within this relatively small area, vegetation experiences a steep gradient of human influence — from the densely cultivated and urbanizing Zarafshan valley to sparsely populated grazing lands at the desert and mountain margins. The province has also undergone considerable change between 2015 and 2024: expansion of irrigated cropland, urban growth around the city of Samarkand, fluctuations in regional drought conditions, and ongoing modernization of the agricultural sector. Disentangling how much of the observed vegetation change reflects climate forcing versus anthropogenic pressure is therefore both scientifically meaningful and policy-relevant.

Conventional single-source assessments are insufficient for this task. Optical vegetation indices alone capture *what* is changing but cannot attribute the change to a driver. Climate datasets alone describe the forcing but not the response. Human-pressure indices describe the disturbance regime but not its ecological impact. A defensible attribution requires the integration of vegetation, climate, and anthropogenic observations through a workflow grounded in sensor physics and statistical reasoning. The present study implements such a workflow for Samarkand Province over the decade 2015–2024.

1.2 Task Definition

The primary objective of this study is to attribute per-pixel trends in the Normalized Difference Vegetation Index (NDVI) over Samarkand Province for the

period 2015–2024 to either climate variability or human land use, and to produce a spatially explicit six-class headline map distinguishing restoration from degradation and identifying the dominant driver in each case. This attribution is required at every grid cell of the analysis domain so that the spatial heterogeneity of climatic and anthropogenic pressures across the province — from the irrigated Zarafshan corridor through urbanizing periphery to semi-arid rangelands — is preserved rather than averaged away [3][4].

Four secondary objectives support this primary aim and structure the analytical workflow.

The first secondary objective is to quantify the temporal trends of every variable that enters the attribution. This is achieved by computing the non-parametric Theil–Sen slope estimator and the Mann–Kendall significance test on the annual time series of NDVI, Land Surface Temperature (LST), precipitation (P), solar radiation (Srad), and the Human Footprint Index (HFI) at every pixel. The resulting maps describe where each variable has changed monotonically and at what rate, and provide the trend terms (dX/dt) used in the subsequent partial-derivative attribution [5]–[7].

The second secondary objective is to assess the explanatory power of a climate-only Ordinary Least Squares (OLS) regression model — the RESTREND framework — at every pixel. The model predicts NDVI from the three climatic drivers (LST, P, Srad), and its per-pixel coefficient of determination (R^2) quantifies the extent to which climatic forcing alone accounts for the observed vegetation variability. The spatial pattern of R^2 identifies where the climate model is robust and, by complement, where the residual signal can be interpreted as non-climatic [1][2].

The third secondary objective is to cross-validate the RESTREND-based attribution using a Random Forest (RF) model that includes HFI as a direct predictor. Whereas RESTREND assumes additive linear relationships between climate variables and NDVI, the Random Forest captures non-linear interactions and threshold effects without imposing a functional form. By training one global RF on the pooled pixel–year dataset and reporting the relative feature importances of LST, P, Srad, and HFI, the study obtains an independent and methodologically distinct ranking of climatic versus anthropogenic explanatory power. Agreement between the two approaches strengthens confidence in the attribution; disagreement is interpreted critically in the Discussion [8][9].

The fourth secondary objective is to validate the non-climatic component of the NDVI trend — the trend of the RESTREND residuals — against the independent trend of the Human Footprint Index. Because the residual trend is derived entirely from satellite vegetation and climate observations, while the HFI trend is derived independently from socio-economic and land-use composites, their per-pixel

correlation constitutes an external consistency check that does not rely on the internal assumptions of the attribution model [10].

The four secondary objectives together produce the evidence base required to answer the primary research question with statistical defensibility, physical interpretability, and an independent cross-check.

1.3 Expected Outputs

The project is designed to produce the following deliverables, each linked directly to one stage of the methodology and to specific evaluation criteria:

A harmonized multi-source data cube covering Samarkand Province at 1 km resolution for 2015–2024, comprising annual stacks of NDVI, LST, precipitation, solar radiation, and Human Footprint Index, all clipped to the AOI and aligned on an identical grid in EPSG:4326. This product demonstrates the integration of optical, thermal, climate-reanalysis, and composite anthropogenic data sources into a single analysis-ready dataset.

A set of per-variable trend maps, including Sen's slope and Mann–Kendall significance for each of the five variables, identifying where vegetation, climate, and human pressure have changed at a statistically meaningful rate over the study decade.

A RESTREND product set, comprising the coefficient maps of the climate regression (sensitivities of NDVI to temperature, precipitation, and solar radiation), the coefficient of determination map indicating where the climate model successfully explains NDVI variance, and the trend of the residuals — the central quantitative expression of human-attributable vegetation change.

A pair of contribution maps quantifying the per-pixel climate contribution (CC) and human contribution (HA) to the observed NDVI trend in NDVI-per-year units, together with their relative-contribution percentages.

A six-class attribution map distinguishing climate-dominated, human-dominated, and co-driven greening from climate-dominated, human-dominated, and co-driven degradation. This map constitutes the headline scientific output of the project and directly answers the research question for every pixel in the AOI.

A Random Forest feature-importance assessment, ranking the relative explanatory power of LST, precipitation, solar radiation, and HFI for NDVI variability across the province, alongside model performance metrics (R^2 , RMSE, MAE) on a spatially held-out test set. This output provides an independent, non-linear cross-validation of the partial-derivative attribution.

A validation scatter plot relating the per-pixel residual trend to the per-pixel Human Footprint trend, with associated correlation statistics, serving as a final

consistency check between two methodologically independent measurements of anthropogenic influence.

A sensitivity analysis testing how the HFI trend, the attribution classes, the Random Forest ranking and the residual–HFI validation change under alternative treatments of the 2018→2019 discontinuity in the Human Footprint record (Sections 3.5.7 and 4.7).

A complete, reproducible processing pipeline implemented in Python, together with this report, that together demonstrate task clarity, logical consistency of workflow, explicit treatment of sensor physics, multi-source integration at three fusion levels (feature, decision, and validation), depth of analysis, and clarity of presentation.

Taken together, these outputs are intended to provide a defensible, physically grounded answer to the research question and to constitute a complete task-driven remote-sensing workflow suitable for the evaluation criteria of the course.

2. Study Area and Data

2.1 Description of the Study Area

Samarkand Province is located in south-central Uzbekistan, spanning approximately 65.14–67.56°E longitude and 39.23–40.68°N latitude and covering roughly 15,100 km² (Figure 1). The province is bisected from east to west by the Zarafshan River, one of the principal rivers of Central Asia, which sustains a narrow but productive irrigated oasis embedded in an otherwise semi-arid to arid landscape. The province occupies a particularly informative position on the climatic and ecological gradient that characterizes Central Asia, ranging from intensive irrigated agriculture along the river corridor to extensive rangeland and foothill steppe on the surrounding loessic uplands. This juxtaposition makes Samarkand well suited to studies that aim to disentangle climatic and anthropogenic drivers of vegetation change.

Study Area: Samarkand Province, Uzbekistan
Mean growing-season NDVI (Apr–Oct, 2015–2024), MODIS MOD13Q1 C6.1

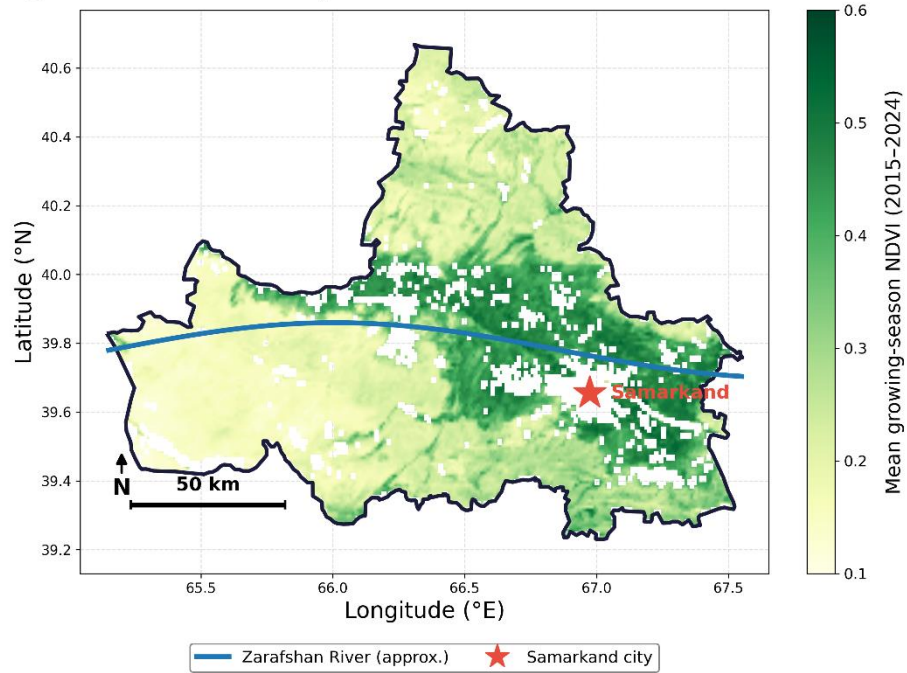


Figure 1. Study area: mean growing-season (April–October) NDVI for 2015–2024, computed from the MOD13Q1 Collection 6.1 stack used in all analyses, with the AOI boundary, the approximate course of the Zarafshan River and the location of Samarkand city.

Physiography. Elevation within the AOI ranges from approximately 270 m in the Zarafshan floodplain to over 2,000 m in the Zarafshan and Turkestan ranges to the northeast. The central and western sections are dominated by loessic foothill steppe and dryland cultivation. The combination of pronounced topographic relief and a strongly continental climate produces a spatially heterogeneous land-surface response to forcing — a feature that motivates per-pixel analysis at 1 km resolution rather than zonal averages.

Climate. The province has a continental semi-arid climate (BSk in the Köppen–Geiger classification). Mean annual precipitation is approximately 300–400 mm in the lowlands, falling predominantly as rainfall and snow between November and April. Growing-season precipitation (April–October) is consequently low, so that vegetation is strongly dependent on irrigation for cropland and on stored soil moisture and groundwater for rangeland. Across the vegetated pixels, mean growing-season daytime Land Surface Temperature (LST) ranges from approximately 21 °C to 48 °C (Table 4). The combination of water-limited rangelands and irrigation-dependent croplands resembles closely the conditions analyzed in published Central Asian studies and makes the region representative of arid-grassland attribution problems more broadly [3].

Land cover (ESA WorldCover 2021, aggregated to 1 km). The AOI comprises 43,309 grid cells at 1 km resolution. After aggregation of the native 10 m WorldCover product to the master grid via majority resampling, the dominant vegetated classes are grassland (28.2 %) and cropland (19.8 %); built-up surfaces, bare ground, and permanent water together account for 3.3 %; tree cover and shrubland are negligible (<0.1 % combined), consistent with the semi-arid character of the province (Table 1). Approximately 48.7 % of the master grid lies outside the AOI polygon and is set to NoData.

***Table 1.** Land-cover composition of the Samarkand AOI on the 1 km master grid (ESA WorldCover 2021 v200, majority resampling). "In mask" indicates classes retained for analysis.*

ESA class	Label	Pixels (1 km)	% of AOI	In mask
0	NoData / outside AOI	21,072	48.7 %	—
10	Tree cover	14	<0.1 %	Yes
20	Shrubland	4	<0.1 %	Yes
30	Grassland	12,212	28.2 %	Yes
40	Cropland	8,555	19.8 %	Yes
50	Built-up	967	2.2 %	—
60	Bare / sparse vegetation	361	0.8 %	—
80	Permanent water	124	0.3 %	—

The vegetation mask retains classes 10, 20, 30, and 40, yielding 20,785 candidate vegetated pixels. After the additional requirement that all five variables (NDVI, LST, P, Srad, HFI) be valid for all ten study years, the working pixel set is reduced to **20,132 valid pixels**, which constitutes the spatial domain for every subsequent analysis. The dominance of grassland and cropland in the mask aligns the present study with the broader literature on arid-grassland and dryland-cropland attribution in Central Asia [3][4] and adjacent regions [9].

2.2 Datasets

Five primary variables and one land-cover ancillary were used in the analysis. All datasets were reprojected to EPSG:4326 and resampled to a common 1 km master grid (pixel size $0.008983^\circ \approx 1,000$ m, derived from the MOD13Q1 NDVI grid), clipped to the Samarkand Province administrative boundary shapefile, and aggregated temporally to growing-season (April–October) annual values over 2015–2024 ($n = 10$ annual observations per pixel). Continuous variables (NDVI, LST, P, Srad, HFI) were resampled using bilinear interpolation; categorical WorldCover was resampled using the majority (mode) rule. Table 2 summarizes the data sources; Sections 2.2.1 – 2.2.6 provide the per-dataset detail.

Table 2. Summary of datasets used in the study. All datasets were harmonized to a 1 km grid in EPSG:4326 clipped to the Samarkand AOI for the period 2015–2024.

Variable	Product	Sensor/ type	Native resolution	Role	Source
NDVI	MODIS MOD13Q1 C6.1	Terra MODIS, optical (R, NIR)	250 m, 16- day	Vegetation response	NASA LP DAA C / GEE
LST	MODIS MOD11A2 C6.1 (LST_Day_1km)	Terra MODIS, thermal IR	~1 km (926 m), 8-day	Climate driver	NASA LP DAAC / GEE
Precipitation (P)	CHIRPS v2.0	Thermal-IR + gauge fusion	0.05° (~5.5 km), monthly totals	Climate driver	UCSB CHC
Solar radiation (Srad)	ERA5-Land SSRD	Reanalysis (model)	0.1° (~11 km), hourly	Climate driver	Copernicus CDS
Human Footprint (HFI)	Mu et al. annual HFI (2019– 2024: later figshare versions; Section 2.2.5)	Composite of 8 anthropogenic layers	~1 km (Mollweide) , annual	Anthropogenic pressure	figshare 16571064, v8
Land cover	ESA WorldCover 2021 v200	Sentinel-1 SAR + Sentinel-2 optical	10 m, single epoch	Vegetation mask	ESA WorldCover

2.2.1 MODIS MOD13Q1 Collection 6.1 — NDVI

NDVI was obtained from the MODIS Terra Vegetation Indices product MOD13Q1, Collection 6.1 (Google Earth Engine collection MODIS/061/MOD13Q1) [13]. MOD13Q1 is a 16-day composite at 250 m native resolution on the MODIS sinusoidal grid. NDVI is computed from surface reflectance in MODIS band 1 (red, 620–670 nm) and band 2 (NIR, 841–876 nm) as $NDVI = \frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + \rho_{Red}}$; the physical basis of the index is discussed in Section 3.3. Within each 16-day window the MODIS vegetation-index algorithm selects the best-quality observation with a constrained-view maximum-value compositing rule, which reduces residual cloud, aerosol and off-nadir effects. The processing applied in this study was, in order: (1) the stored integers were converted to NDVI with the product

scale factor 0.0001 (valid range -0.2 to 1.0); (2) no additional per-pixel screening with the SummaryQA, DetailedQA or pixel-reliability layers was applied, so the analysis relies on the quality selection of the compositing algorithm (a limitation discussed in Section 5.2); (3) for each calendar month, all 16-day composites whose start date falls in that month (one or two per month) were averaged to a monthly mean NDVI; (4) the monthly images were exported from Google Earth Engine [18] in EPSG:4326 at a nominal 1 km scale (0.008983°), which defines the master grid (Section 3.2.1), so the 250 m values were aggregated to 1 km by Earth Engine's default resampling at export, without an explicit `reduceResolution` step; and (5) growing-season annual NDVI was computed as the NaN-aware mean of the seven monthly values from April to October (Section 3.2.3). The 120 monthly files cover January 2015 to December 2024. Figure 1 and every NDVI result in this report are derived from this single MOD13Q1 Collection 6.1 stack.

2.2.2 MODIS MOD11A2 Collection 6.1 — Land Surface Temperature

Daytime Land Surface Temperature was obtained from MODIS MOD11A2, Collection 6.1 (LST_Day_1km; Google Earth Engine collection MODIS/061/MOD11A2) [14], an 8-day composite of daily retrievals at ~ 1 km (926 m) native resolution. LST is retrieved from thermal-infrared bands 31 ($10.78\text{--}11.28\ \mu\text{m}$) and 32 ($11.77\text{--}12.27\ \mu\text{m}$) with the generalized split-window algorithm, using classification-based emissivities; the physical basis (Planck's law) is outlined in Section 3.3. Stored values were converted to kelvin with the scale factor 0.02 and then to $^\circ\text{C}$. Pixels without a valid retrieval (for example, persistently cloudy pixels) are already masked in the product; no additional filtering with the QC_Day bit field was applied. Monthly LST was computed as the mean of the 8-day composites starting in each month, exported at the same 1 km scale as NDVI, and averaged over April–October in the same way. Daytime LST is widely used as a covariate in RESTREND-type attribution because it integrates the surface energy balance and is spectrally independent of NDVI [3].

2.2.3 CHIRPS v2.0 — Precipitation

Precipitation was obtained from the Climate Hazards group InfraRed Precipitation with Stations (CHIRPS) v2.0 dataset [15] produced by the UC Santa Barbara Climate Hazards Center. CHIRPS fuses satellite Cold Cloud Duration (CCD) estimates derived from geostationary thermal-infrared observations with a Climate Hazards group Precipitation climatology and station gauge records via a modified inverse-distance interpolation; it therefore constitutes an internal multi-source fusion

product in its own right. The native resolution is 0.05° (~ 5.5 km), and the dataset is provided at daily, pentadal and monthly time steps. Monthly precipitation totals were used in this study and resampled to the 1 km master grid by bilinear interpolation; the 120 monthly files cover 2015–2024. CHIRPS has been widely adopted for dryland precipitation analysis in Central Asia and Inner Asia because of its long record, station correction, and suitability for data-sparse regions [4][9].

2.2.4 ERA5-Land — Surface Solar Radiation Downwards (Srad)

Surface Solar Radiation Downwards (SSRD) was obtained from the ERA5-Land reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and distributed through the Copernicus Climate Data Store [16]. ERA5-Land is an offline land-surface replay of the ERA5 atmospheric reanalysis at finer spatial resolution ($0.1^\circ \approx 11$ km, hourly); SSRD is a forecast accumulation expressed in J m^{-2} . Hourly accumulations were converted to mean flux and aggregated to monthly means in W m^{-2} (the unit used throughout this report; Table 4), and resampled to the 1 km master grid. SSRD acts as a proxy for photosynthetically active radiation (PAR), which controls the energy available for primary production.

2.2.5 Global Human Footprint Index (HFI)

The Human Footprint Index was obtained from the annual global Human Footprint dataset of Mu et al. [10], distributed through figshare [11]. HFI is a composite anthropogenic-pressure index that combines eight weighted pressure layers — built environments, population density, night-time lights, croplands, pastures, roads, railways and navigable waterways — into a dimensionless score from 0 to 50, following the weighting scheme of Venter et al. [19]. Four of the eight inputs (built-up area, night-time lights, croplands and pastures) are derived from satellite imagery, so the index has a substantial remote-sensing component, although it is not a direct observation.

Temporal coverage and provenance of the 2019–2024 layers. The peer-reviewed description of the dataset (Mu et al., 2022) covers 2000–2018 only. The HFI rasters for 2019–2024 used here were not extrapolated, reconstructed from individual components, taken from another dataset or copied from 2018. They were downloaded, together with the 2015–2018 layers, from the same figshare record, which the authors have since extended: version 8 of the record (last modified 5 November 2025 and still titled “...from 2000 to 2018”) contains one file per year from hfp2000 to hfp2024. How the 2019–2024 layers were produced — in particular

which input layers were updated and whether the processing chain is identical to that of the 2000–2018 release — is not documented in the paper or in the record description. These layers are therefore treated as a provisional extension, and three checks were made:

- The 2019–2024 rasters are not duplicates of 2018: no valid pixel has the same value in 2018 and 2019.
- The series contains a marked discontinuity between 2018 and 2019 (Figure 9): the province-mean HFI of the analysed pixels rises from 14.23 to 16.85 (mean per-pixel change +2.62, against a mean absolute change of 0.49 for the other eight year-to-year steps); 90.2% of pixels increase, and for 46.2% of pixels this single step is the largest annual change of the decade. The jump is concentrated in the cropland-dominated centre and east (+3.9 to +5.8 index units in the zones east of 66.6°E) and is small in the western rangelands (+0.3 to +0.6).
- Between 2021 and 2024, 25–39% of pixels have exactly the same value as in the previous year, which suggests that some component layers are not updated every year. A step of this size is more likely to reflect a change in inputs or processing than a real one-year change in human pressure, so the robustness of all HFI-dependent results to this break is tested explicitly (Sections 3.5.7 and 4.7).

The annual rasters, distributed at ~1 km in the Mollweide projection and supplied for this study clipped to Samarkand in EPSG:4326 (0.00989°, ~1,101 m), were resampled bilinearly to the master grid; the ten annual files cover 2015–2024. Annual human-footprint maps of this kind are increasingly used to characterize changing human pressure in Central and High Asia [29].

2.2.6 ESA WorldCover 2021

The land-cover ancillary was obtained from ESA WorldCover 2021 v200, a global 10 m land-cover product derived from a fusion of Sentinel-1 C-band SAR backscatter and Sentinel-2 multispectral optical observations. The product distinguishes 11 land-cover classes following a modified UN Land Cover Classification System. The native 10 m raster was aggregated to the 1 km master grid using majority (mode) resampling — appropriate for categorical data because numerical interpolation between class codes is not meaningful.

2.3 Justification of the Dataset Combination

The combination of MOD13Q1 NDVI, MOD11A2 LST, CHIRPS, ERA5-Land Srad, the Mu et al. HFI, and ESA WorldCover is not arbitrary; each dataset

fills a specific role in the task-driven framework that no other product fills as well over the study domain and period.

The two MODIS products provide spectrally independent observations of *vegetation state* (NDVI from reflected solar radiation in the visible and NIR) and *thermal state* (LST from emitted thermal-infrared radiation), thereby separating biophysical greenness from energy balance. CHIRPS introduces *water availability*, the dominant ecological forcing in semi-arid Central Asia, with the additional methodological strength that it is itself a satellite–gauge fusion product. ERA5-Land Srad introduces *energy availability* (PAR proxy), completing the standard climatic triplet (T, P, Srad) used in the partial-derivative attribution framework of comparable published studies [3][4][12][24]. HFI provides the *anthropogenic-pressure* axis without which climate-only RESTREND cannot be turned into a true attribution; its annual cadence and 1 km grid match the analytical needs of the partial-derivative and Random Forest stages described in Section 3.5. WorldCover restricts the analysis to *valid vegetated surfaces*, ensuring physical interpretability of NDVI residuals.

Functionally, the six products represent four sensor-physics families: passive optical (MOD13Q1, S2 component of WorldCover), passive thermal-infrared (MOD11A2, and indirectly CHIRPS through CCD), active microwave (S1 component of WorldCover), and reanalysis-based atmospheric forcing (ERA5-Land). This multi-sensor composition directly supports the assignment's emphasis on multi-source integration and sensor-physics understanding (Section 3.3 and 3.4). The integration of independently produced datasets at three fusion levels — feature-level (regression of NDVI on T, P, Srad), decision-level (six-class attribution map), and validation-level (independent cross-check of RESTREND residual trend against HFI trend) — is detailed in Section 3.4.

3. Methodology

3.1 Overall Workflow

The analytical pipeline follows the task-driven framework prescribed by the assignment, in which a defined application problem (per-pixel attribution of NDVI trends to climatic versus anthropogenic drivers in Samarkand Province, 2015–2024) determines the choice of data sources, preprocessing, features, fusion strategy, and analytical methods. The pipeline comprises seven explicit stages, executed in strict sequence and illustrated in Figure 2.

Stage 1 — Data acquisition. Six datasets were acquired: MODIS MOD13Q1 C6.1 NDVI and MOD11A2 C6.1 LST (via Google Earth Engine), CHIRPS v2.0 precipitation, ERA5-Land SSRD, the Mu et al. annual Human Footprint Index (2015–2018 from the peer-reviewed release; 2019–2024 from the extended figshare record, Section 2.2.5) and ESA WorldCover 2021. The five primary variables cover 2015–2024; WorldCover is a single 2021 epoch [10][11][13]–[18].

Stage 2 — Spatial harmonization. Every raster is reprojected to EPSG:4326 and resampled to a single 1 km master grid (269×161 pixels) derived from the MOD13Q1 NDVI tile for April 2015. Continuous variables (NDVI, LST, precipitation, SRAD, HFI) are resampled by bilinear interpolation; the categorical WorldCover product is aggregated by majority (mode) resampling. All rasters are clipped to the Samarkand administrative boundary.

Stage 3 — Temporal aggregation. Monthly NDVI, LST, precipitation, and SRAD rasters are aggregated to growing-season (April–October) annual means using NaN-aware reduction. HFI is retained at its native annual cadence. This produces five three-dimensional arrays of shape (10 years, 161 rows, 269 columns), with $n = 10$ annual observations per pixel [1][2].

Stage 4 — Masking and quality control. A binary vegetation mask is built from ESA WorldCover by retaining classes 10 (tree), 20 (shrubland), 30 (grassland), and 40 (cropland). The mask is intersected with a pixel-wise completeness requirement: a pixel is retained only if it has valid (non-NaN) values in all five variables across all ten years. The intersection yields 20,132 valid pixels (46.5 % of the AOI grid; 96.9 % of vegetated pixels). All subsequent analysis operates on this fixed pixel set.

Stage 5 — Trend feature extraction. For every valid pixel and every variable, the non-parametric Theil–Sen slope estimator and the Mann–Kendall trend test are applied to the 10-year time series. This produces ten output rasters: per-pixel slope and significance for NDVI, LST, precipitation, SRAD, and HFI.

Stage 6 — RESTREND climate regression and partial-derivative attribution. A per-pixel ordinary-least-squares (OLS) regression $\text{NDVI} \sim a \cdot T + b \cdot P + c \cdot \text{Srad} + \delta$ is fitted on the annual stacks. The coefficients (a , b , c) are the per-pixel sensitivities of NDVI to the three climate drivers; their multiplication by the corresponding Sen slopes (dT/dt , dP/dt , $d\text{Srad}/dt$) yields the Climate Contribution map (CC). Residuals from the climate regression are regressed against annual HFI to yield the HFI sensitivity e , which multiplied by $d\text{HFI}/dt$ yields the Human Attribution map (HA). The signs and relative magnitudes of CC and HA, combined with the sign of the observed NDVI Sen slope, classify each pixel into one of six attribution categories (climate-/human-/co-driven restoration or degradation) [1][3].

Stage 7 — Independent cross-validation and sensitivity analysis. Two methodologically distinct checks are performed. First, a Random Forest regressor is trained on the pooled pixel–year dataset ($\text{NDVI} \sim \text{LST} + \text{P} + \text{Srad} + \text{HFI} + \text{lat} + \text{lon} + \text{year}$) with a spatial 80/20 train–test split; impurity-based (Mean Decrease Impurity) feature importances rank the explanatory power of the four physical drivers without imposing a linear-additive form. Second, the per-pixel residual Sen slope from the RESTREND model is correlated with the per-pixel HFI Sen slope (Pearson and Spearman) to test whether the non-climatic component of NDVI change covaries spatially with independently measured anthropogenic pressure. Both checks, and all HFI-dependent outputs, are repeated under alternative treatments of the 2018→2019 HFI discontinuity (Section 3.5.7) [8][25].

The pipeline implements three distinct levels of multi-source data fusion. Feature-level fusion occurs in Stage 6, where heterogeneous datasets (optical NDVI, thermal LST, satellite-gauge precipitation, reanalysis SRAD, composite HFI) are integrated as predictors in a single per-pixel model. Decision-level fusion occurs in the production of the six-class attribution map, where continuous CC and HA outputs are combined with the sign of the NDVI trend to assign each pixel a discrete categorical label. Validation-level fusion occurs in Stage 7, where two independently produced anthropogenic-pressure measurements (the RESTREND residual trend and the HFI trend) are cross-compared.

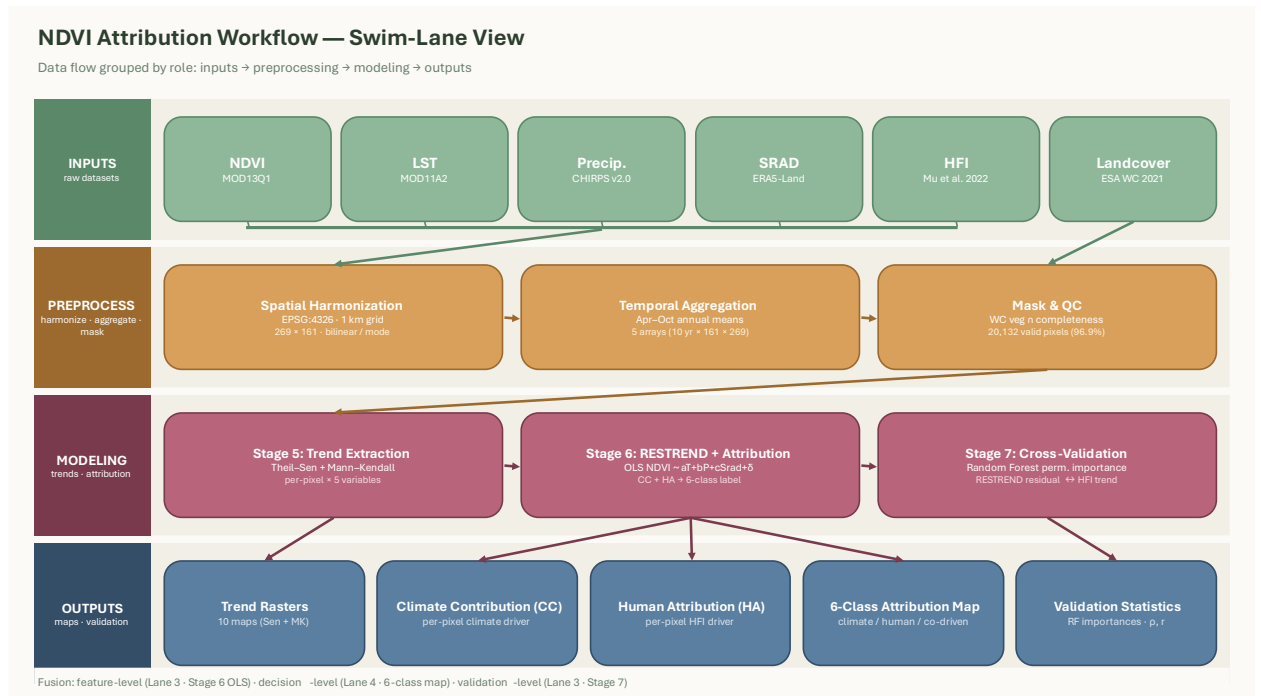


Figure 2. Overall workflow of the multi-source attribution pipeline. Inputs are harmonized to a common 1 km grid, aggregated temporally, masked and quality-controlled, then passed through parallel analytical branches: per-pixel trend analysis, RESTREND + partial-derivative

attribution (producing the six-class attribution map), and Random Forest + residual-vs-HFI validation.

3.2 Preprocessing

3.2.1 Master Grid Definition

A single reference grid was derived from the MOD13Q1 NDVI tile for April 2015 and adopted as the master coordinate system for all subsequent processing. The grid has the following properties:

Table 3. *Properties of the 1 km master grid shared by all harmonized rasters.*

Property	Value
Coordinate Reference System	EPSG:4326 (WGS 84 geographic)
Origin (top-left corner)	65.1368° E, 40.6757° N
Pixel size	0.008983° ($\approx 1,000$ m at the equator; ≈ 760 m N–S $\times \approx 850$ m E–W at 40° N)
Grid dimensions	269 columns $\times$ 161 rows = 43,309 cells
Spatial extent	Samarkand Province, Uzbekistan

EPSG:4326 was chosen for interoperability: CHIRPS, ERA5-Land and WorldCover are distributed in geographic coordinates, and the MODIS (sinusoidal) and HFI (Mollweide) products were reprojected to it on export, giving a single consistent reference for processing in Python (rasterio) and visualization in QGIS. Because geographic pixels are not equal-area, pixel areas are computed at the grid-centre latitude (0.762 km²). The 1 km pixel size matches the native resolution of MOD11A2 LST and of the HFI product (Section 2.2) and is the coarsest meaningful scale of the analysis, given that CHIRPS (~ 5.5 km) and ERA5-Land (~ 11 km) are necessarily resampled to participate in pixel-level fusion.

3.2.2 Raster Alignment and Reprojection

Every input raster was reprojected and resampled onto the master grid using GDAL warp (via rasterio.warp.reproject) with bilinear interpolation for continuous variables and nearest-neighbour / majority resampling for the categorical land-cover product. Three dataset-specific corrections were required:

- **ERA5-Land SSRD.** The native ERA5-Land grid is offset by approximately 0.004° westward relative to the MOD13Q1 grid; the reproject step snapped each pixel to the master origin. Hourly ERA5-Land values were aggregated to monthly means (W m^{-2}) before harmonization.
- **HFI.** The Mu et al. HFI is distributed at 1 km in the Mollweide equal-area projection; the clipped annual rasters used here had been reprojected to EPSG:4326 at 0.00989° ($\sim 1,101$ m) on a 244×146 grid. They were resampled bilinearly to the 1,000 m master grid, and the tagged NoData value ($-\infty$) was converted to NaN to ensure consistent missing-value handling across the stack.
- **ESA WorldCover 2021.** The native 10 m raster ($26,838 \times 16,011$ pixels for the AOI) was aggregated to 1 km using majority (mode) resampling. This preserves categorical integrity, which would be destroyed by numerical interpolation between discrete class codes.

After alignment, every output raster was verified to share identical transform, dimensions, CRS, and NoData definition. Verification passed for all 481 rasters in the harmonized stack (120 monthly NDVI + 120 monthly LST + 120 monthly CHIRPS + 120 monthly SRAD + 10 annual HFI + 1 WorldCover).

3.2.3 Growing-Season Annual Aggregation

Vegetation in Samarkand Province is dormant or snow-covered outside the warm season, and including these months would inject noise unrelated to vegetation response to climate. The growing season was therefore defined as April through October (months 4–10), consistent with prior dryland and Central Asian studies [3][4]. For each year between 2015 and 2024 and each pixel, the seven monthly aligned files were aggregated into a single annual value:

- **NDVI, LST, SRAD** — pixel-wise mean across the seven growing-season months (`np.nanmean`), preserving valid observations where at least one month per year is non-missing.
- **Precipitation (CHIRPS)** — pixel-wise mean of monthly totals (mm month^{-1}), retained in the same units as the input for direct comparability with the trend analysis.
- **HFI** — already provided as a single annual composite per year; loaded directly without further aggregation.

The use of `np.nanmean` rather than a plain mean is methodologically important: it ensures that a single cloud-contaminated month (NaN in NDVI) does not invalidate the entire annual aggregate, while still preserving NaN where all seven months are missing. The resulting arrays have shape (10 years, 161 rows, 269 columns) for each of the five variables.

3.2.4 Vegetation Masking and Valid-Pixel Selection

Two filters were applied to identify the analytical pixel set on which all subsequent computations operate.

The **vegetation mask** was derived from ESA WorldCover 2021 by retaining pixels labelled as tree cover (class 10), shrubland (20), grassland (30), or cropland (40). The mask comprises 20,785 pixels (48.0 % of the AOI grid). Built-up surfaces (class 50), bare or sparse vegetation (60), permanent water (80), and outside-AOI cells (class 0) were excluded. This step prevents spectral contamination of NDVI by non-vegetated surfaces and ensures that the residual term in RESTREND can be physically interpreted as a vegetation response.

The **completeness filter** required every retained pixel to have valid, non-NaN observations in all five variables for all ten years. Pixels failing this requirement were removed from the analytical set. The retained pixel set is dominated by grassland and cropland, in proportions broadly consistent with the AOI-wide land-cover composition reported in Section 2.1, and it is held fixed across all ten years of the analysis. This is essential for trend analysis: a pixel set that varied year-to-year would conflate temporal and sampling variability.

3.2.5 Harmonized Data Cube

All values fall within physically plausible ranges for a semi-arid Central Asian province. The mean growing-season NDVI of 0.28 is consistent with the predominance of sparsely vegetated steppe and dryland cropland; the mean growing-season LST of 36.8 °C reflects the strong continental insolation regime; mean precipitation of $\sim 13 \text{ mm month}^{-1}$ ($\approx 90 \text{ mm}$ over the seven-month growing season) confirms severe water limitation; and the HFI distribution (mean 15.5, max 46.2) captures the strong contrast between rangeland with minimal pressure and intensely modified urban and irrigated lowlands.

Table 4. Sanity statistics for the harmonized data cube, computed over the 20,132 valid pixels and 10 years ($n = 201,320$ pixel-years).

Variable	Min	Max	Mean	Std. dev.	Units
NDVI	0.024	0.630	0.279	0.121	dimensionless
LST	20.82	47.86	36.76	4.91	°C
Precipitation	1.76	49.23	13.32	6.87	mm month ⁻¹
SRAD	248.3	278.0	263.6	6.17	W m ⁻²
HFI	5.41	46.21	15.52	7.56	index (0–50)

Interannual NDVI variability over the study period (Table 5) shows a marked minimum in 2018 (mean NDVI 0.232) and a maximum in 2024 (0.314). The 2018 minimum is consistent with documented drought conditions in Central Asia during that year and provides an a-priori reasonableness check on the data cube: a known regional climatic anomaly is correctly reflected in the harmonized vegetation signal. The values are otherwise stable, with all ten years comprising exactly 20,132 valid pixels — confirming that the spatial mask was applied identically across the time stack and that no pixel-set drift contaminates the subsequent trend analysis.

Table 5. Interannual variability of growing-season mean NDVI across the 20,132 valid pixels.

Year	Mean NDVI	Std. dev.
2015	0.286	0.104
2016	0.283	0.115
2017	0.275	0.113
2018	0.232	0.126 — lowest
2019	0.308	0.114
2020	0.311	0.112
2021	0.249	0.116
2022	0.276	0.125
2023	0.252	0.130
2024	0.314	0.121 — highest

3.3 Feature Extraction

For each variable, a single per-pixel annual value was extracted from the harmonized data cube (Section 3.2). NDVI, LST, precipitation, and SRAD were averaged across the seven growing-season months (April–October) of each year; HFI was used at its native annual cadence. This produced five three-dimensional arrays of shape (10 years, 161 rows, 269 columns) — the inputs to all subsequent trend and attribution analyses.

NDVI is the response variable. As defined in Section 2.2.1, it is computed from the red/NIR reflectance contrast and is interpreted here as a measure of canopy chlorophyll content and photosynthetic capacity. The annual growing-season mean was chosen rather than a peak or instantaneous value because vegetation in semi-arid Samarkand responds to *cumulative* seasonal forcing, not to short-term anomalies. Observed values across the 20,132 valid pixels span 0.024–0.630, comfortably below the NDVI saturation threshold of ~0.7–0.8.

LST is the temperature predictor. The MOD11A2 daytime retrieval (Section 2.2.2) represents surface radiative temperature at the Terra overpass (~10:30 LST) and integrates the surface energy balance. The growing-season mean captures cumulative heat exposure during the active vegetation period — the physically relevant aggregation for evapotranspiration and thermal stress on dryland vegetation.

Precipitation is the water-availability predictor. CHIRPS monthly totals (Section 2.2.3) were averaged across the growing season to yield a single $\text{mm}\cdot\text{month}^{-1}$ value per year. Since CHIRPS' known winter snow-undercount falls outside the April–October window, the growing-season aggregation also removes the most significant residual bias in the product over Central Asia [15].

SRAD is the radiation predictor. ERA5-Land hourly SSRD (Section 2.2.4) was aggregated to monthly means and then growing-season means in $\text{W}\cdot\text{m}^{-2}$, providing a proxy for the photosynthetically active radiation available to vegetation.

HFI is the anthropogenic predictor. The annual Mu et al. composite (including the 2019–2024 extension) (Section 2.2.5) was used directly without temporal aggregation; the underlying physical mechanism is the combined pressure from built-up area, population density, nighttime lights, croplands, pastures, and transportation networks.

ESA WorldCover is used not as a feature but as a vegetation mask: pixels not classified as tree, shrub, grassland, or cropland in 2021 are excluded from all analyses (Section 3.2.4).

No higher-order features (texture, time-lag, spectral indices beyond NDVI) were extracted. The deliberately minimal feature set — one biophysical response variable, three climatic drivers, one anthropogenic driver — keeps the per-pixel

regression well-conditioned for the $n = 10$ sample size and avoids the overfitting risks that more elaborate feature engineering would introduce.

3.4 Data Fusion Strategy

The pipeline implements three distinct levels of multi-source fusion, in addition to the spatial harmonization that brings all rasters onto a common 1 km grid. The three fusion levels operate at different conceptual stages of the workflow and serve complementary roles.

Feature-level fusion is performed in Stage 6 (RESTREND), where heterogeneous datasets — optical NDVI, thermal LST, satellite-gauge CHIRPS precipitation, reanalysis ERA5-Land SSRD, and the composite Human Footprint Index — are integrated as predictors in a single per-pixel ordinary-least-squares regression and a subsequent residual-vs-HFI regression. Each input contributes a physically distinct piece of information (vegetation greenness, surface energy state, water input, radiation forcing, anthropogenic pressure); their joint use in a multivariate model is the mechanism by which the climate signal is isolated from the anthropogenic signal in NDVI. This level is appropriate because the inputs are physically complementary but operate on the same target variable (NDVI), and they vary on commensurate annual time scales after Stage 3.

Decision-level fusion is performed in the production of the six-class attribution map (Stage 6 output). Three continuous variables produced by feature-level fusion — the observed NDVI Sen slope, the Climate Contribution (CC), and the Human Attribution (HA) — are combined with discrete decision rules (signs and similarity ratio) to assign every pixel a single categorical label drawn from {climate-, human-, co-driven} $\times$ {restoration, degradation}. Decision-level fusion is appropriate at this stage because the analytical question becomes categorical (which driver dominates?), and discrete classes are far more interpretable for managers and reviewers than continuous CC and HA rasters in isolation.

Validation-level fusion is performed in Stage 7, where two independently produced anthropogenic measurements are cross-compared. The RESTREND residual Sen slope is derived entirely from satellite NDVI and the three climate variables; the HFI Sen slope is derived independently from a separate socioeconomic-and-satellite composite. Their per-pixel correlation, together with the agreement between the partial-derivative attribution and the Random Forest feature-importance ranking, constitutes a multi-method cross-check that does not depend on the internal assumptions of either single method. This level is appropriate because it tests whether the inferred human signal is consistent with an independent

observation of human pressure — the strongest possible form of internal validation given the available data.

Data-level fusion (i.e., direct pixel-stacking before any analysis) is not used in this study because the input datasets differ in native sensor type, retrieval algorithm, and spectral content, making meaningful direct combination at the radiance/reflectance level scientifically inappropriate. Spatial harmonization to a common grid (Section 3.2.2) is a prerequisite for, but distinct from, fusion.

3.5 Analytical Method

3.5.1 Theil–Sen Slope and Mann–Kendall Test

For each pixel time series (t_i, y_i) with $n = 10$ annual observations, the Theil–Sen slope estimator is computed as the median of all $C(10,2) = 45$ pairwise slopes:

$$\beta_{\text{Sen}} = \text{median}_{i < j} \left(\frac{y_j - y_i}{t_j - t_i} \right)$$

This estimator is robust to outliers (a single anomalous year cannot dominate the median) and makes no distributional assumption; the Mann–Kendall statistic, being rank-based, is in addition invariant under monotonic transformations of the data. Statistical significance is tested with the non-parametric Mann–Kendall test:

$$S = \sum_{i < j} \text{sgn}(y_j - y_i)$$

$$\text{Var}(S) = \frac{n(n-1)(2n+5)}{18}$$

The standardized statistic $Z = \frac{S - \text{sgn}(S)}{\sqrt{\text{Var}(S)}}$ is approximately normal under the null hypothesis of no trend; the two-sided p-value is $p = 2[1 - \Phi(|Z|)]$. For $n = 10$, attaining $p < 0.05$ requires $|\tau| > 0.51$, a high threshold that explains the relatively low percentage of individually significant climate-variable pixels observed in the Results section. Implementation uses *pymannkendall.original_test()* applied per pixel, parallelized row-by-row with *joblib.Parallel(n_jobs = -1)* and *tqdm* progress tracking [5]–[7][20][21]. $\text{Var}(S)$ includes the standard correction for tied values, which matters for HFI because some pixels repeat identical values in consecutive years.

3.5.2 RESTREND — Per-Pixel Climate Regression

For each valid pixel p , annual NDVI is regressed on the three climate predictors across $n = 10$ years:

$$NDVI_{p,t} = \beta_{T,p} LST_{p,t} + \beta_{P,p} P_{p,t} + \beta_{S,p} SRAD_{p,t} + \delta_p + \varepsilon_{p,t}$$

The system is solved by ordinary least squares via the normal equations. Defining $X_p \in \mathbb{R}^{10 \times 4}$ (LST, Precip, SRAD, ones) and $y_p \in \mathbb{R}^{10}$ (NDVI):

$$\hat{\beta}_p = (X_p^T X_p)^{-1} X_p^T y_p$$

Implementation uses a fully vectorized solver. The $X^T X$ tensor of shape $(n_{\text{valid}}, 4, 4)$ is built via *numpy.einsum*, inverted by *numpy.linalg.pinv* (which handles near-singular pixels by truncated SVD), and applied simultaneously to all 20,132 pixels.

Per-pixel goodness of fit is reported as both R^2 and adjusted R^2 :

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

$$\begin{aligned} R_{\text{adj}}^2 &= 1 - (1 - R^2) \cdot \frac{n - 1}{n - k - 1} \\ &= 1 - (1 - R^2) \cdot \frac{9}{6} \end{aligned}$$

with $k = 3$ predictors and $v = n - k - 1 = 6$ residual degrees of freedom. Adjusted R^2 is the meaningful metric here because with only 6 dof the raw R^2 is upward-biased; adjusted R^2 penalizes the fit for the number of predictors. The RESTREND residual $\hat{\varepsilon}_{p,t} = NDVI_{p,t} - \widehat{NDVI}_{p,t}$ is retained and submitted to a per-pixel Theil–Sen / Mann–Kendall analysis identical to that of Section 3.5.1, producing the per-pixel residual Sen slope and significance map [1][22][23].

3.5.3 Partial-Derivative Attribution

The Climate Contribution to the observed NDVI trend at pixel p is obtained by multiplying each climate coefficient (interpreted as a partial derivative $\partial NDVI / \partial X$) by the corresponding climate Sen slope:

$$CC_p = \hat{\beta}_{T,p} \cdot \beta_{LST,p}^{\text{Sen}} + \hat{\beta}_{P,p} \cdot \beta_{P,p}^{\text{Sen}} + \hat{\beta}_{S,p} \cdot \beta_{SRAD,p}^{\text{Sen}}$$

For the human signal, a second per-pixel OLS is fitted between the RESTREND residuals and annual HFI:

$$\hat{\varepsilon}_{p,t} = e_p \cdot HFI_{p,t} + c_p + \eta_{p,t}$$

yielding the HFI sensitivity e_p , and the Human Attribution:

$$HA_p = e_p \cdot \beta_{HFI,p}^{\text{Sen}}$$

Relative contributions are computed as [24]:

$$CC\% = \frac{|CC|}{|CC| + |HA|} \times 100,$$

$$HA\% = \frac{|HA|}{|CC| + |HA|} \times 100$$

3.5.4 Six-Class Attribution Rule

Each pixel is assigned a class label by applying the following priority-ordered rule. Define the similarity ratio $\rho_p = \frac{|CC_p - HA_p|}{\max(|CC_p|, |HA_p|)}$.

- Class 3 — co-driven restoration: NDVI slope > 0 and $\rho_p < 0.2$
- Class 6 — co-driven degradation: NDVI slope < 0 and $CC < 0$ and $HA < 0$ and $\rho_p < 0.2$
- Class 1 — climate-dominated restoration: NDVI slope > 0 , $CC > 0$, $HA > 0$, $CC > HA$
- Class 2 — human-dominated restoration: NDVI slope > 0 , $CC > 0$, $HA > 0$, $HA \geq CC$
- Class 4 — climate-dominated degradation: NDVI slope < 0 , $CC < 0$, $|CC| > |HA|$
- Class 5 — human-dominated degradation: NDVI slope < 0 , $HA < 0$, $|HA| > |CC|$
- Class 0 — unclassified: any other configuration (mixed-sign or near-zero contributions)

Co-driven classes are evaluated first to prevent them being mis-labelled as dominant-driver classes.

3.5.5 Random Forest Cross-Validation

A single global Random Forest regressor was trained on the pooled pixel–year dataset to provide an independent, non-linear cross-check of the linear RESTREND attribution. The model predicts annual NDVI from four physical drivers and three nuisance controls:

$$NDVI_{p,t} = f(LST, P, SRAD, HFI, lat, lon, year)$$

The ensemble consists of $B = 300$ regression trees (CART), each grown to a maximum depth of 12 on a bootstrap sample of the training data, with all seven features evaluated at every split (the scikit-learn default for regression, `max_features = 1.0`). Hyper-parameters: `n_estimators = 300`, `max_depth = 12`, `n_jobs = -1`, `random_state = 42`.

A spatial 80/20 hold-out split was implemented by hashing each pixel's (lat, lon) pair with MD5, mapping the hash to a bucket in $[0, 100)$, and assigning pixels with bucket < 80 to the training set and the remainder to the test set. All 10 years of a given pixel are kept in the same split, eliminating temporal leakage. Feature importance is measured as Mean Decrease Impurity (Gini importance) — the variance-reduction contributed by each feature averaged across all 300 trees; because MDI can favour continuous predictors with many possible split points [26], it is interpreted as a ranking rather than an exact partition of explained variance. Model fit on the held-out test set is reported as $R^2 = 1 - \frac{\sum(y-\hat{y})^2}{\sum(y-\bar{y})^2}$ [8][25].

3.5.6 Independent Residual–HFI Validation

The final validation regresses the per-pixel RESTREND residual Sen slope against the per-pixel HFI Sen slope using Pearson and Spearman correlations and ordinary least squares. This step tests whether the spatial pattern of non-climatic NDVI change covaries with the spatial pattern of independently measured anthropogenic pressure, providing a methodologically independent check on the attribution direction [1][29]. Its robustness to the HFI discontinuity is assessed in Section 4.7.

3.5.7 Sensitivity to the HFI Discontinuity

Because the 2019–2024 HFI layers come from an undocumented extension of the Mu et al. record and contain a step change between 2018 and 2019 (Section 2.2.5), every HFI-dependent result was recomputed under two alternative treatments of the series, keeping NDVI, the climate variables, the RESTREND residuals and

CC unchanged. In scenario S1 (step-adjusted), the 2019–2024 values of each pixel are shifted by that pixel's 2018→2019 change, i.e. the step is assumed to be entirely artificial and real change resumes from 2019. In scenario S2 (frozen), the 2018 value is repeated for 2019–2024, i.e. HFI is treated as unknown after the last documented year. For each scenario the HFI Sen slope and Mann–Kendall test, the coefficient e , HA, the six-class map, the residual–HFI correlation and the Random Forest (identical hyper-parameters and spatial split) were recomputed and compared with the original analysis (S0). As a further diagnostic, the per-pixel significance of the residual–HFI regression and the temporal variability of HFI were examined within each attribution class.

4. Results

4.1 Per-Variable Trend Maps

Per-pixel Theil–Sen slopes and Mann–Kendall significance were computed for all five variables across the 20,132 valid pixels and ten years (Table 6).

Table 6. Per-variable Sen's slope statistics and Mann–Kendall significance.

Variable	Unit	Mean slope	Median slope	% pixels $p < 0.05$
NDVI	NDVI yr ⁻¹	+0.00016	−0.00036	6.4%
LST	K yr ⁻¹	−0.165	−0.158	3.0%
Precipitation	mm month ⁻¹ yr ⁻¹	+0.196	+0.201	0.0%
SRAD	W m ⁻² yr ⁻¹	+0.305	+0.274	0.0%
HFI	index yr ⁻¹	+0.198	+0.027	39.9%

The fraction of individually significant pixels is low for NDVI, LST, precipitation, and SRAD. This is the expected consequence of the limited Mann–Kendall power at $n = 10$, where $|\tau| > 0.51$ is required for $p < 0.05$ (Section 3.5.1). HFI is the only variable with substantial significant coverage (39.9%: 22.6% increasing and 17.3% decreasing), partly because the 2018→2019 step (Section 4.7) splits many series into two nearly constant levels, which the rank-based test registers as a monotonic trend.

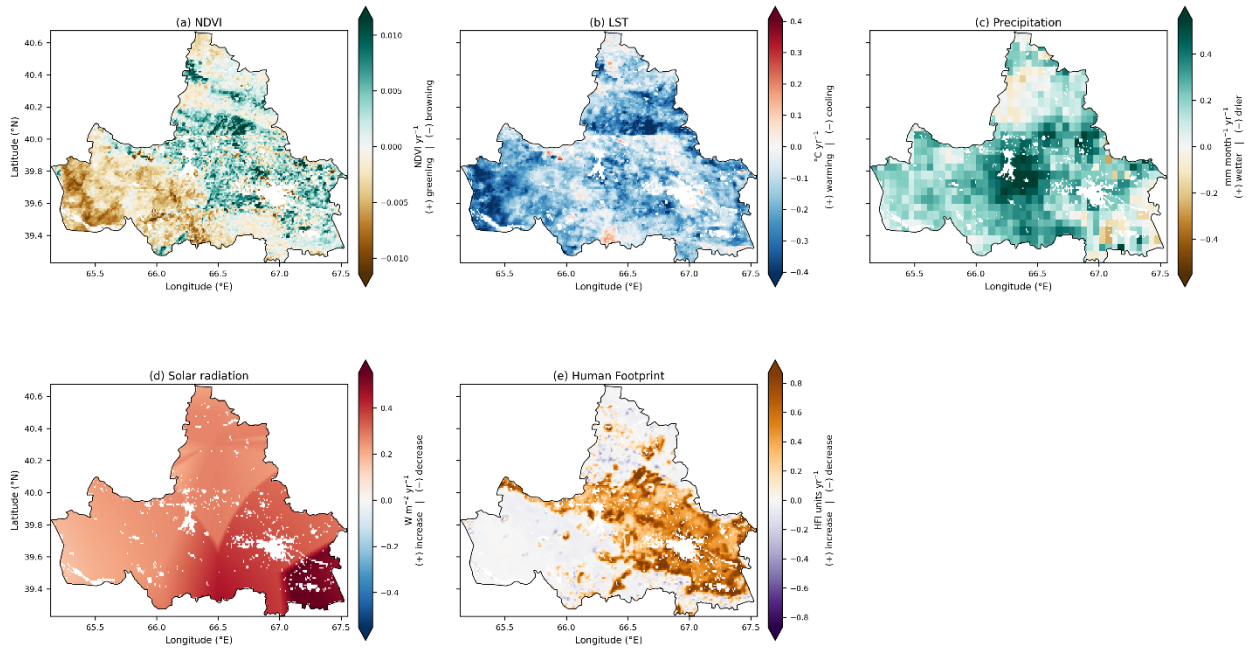


Figure 3. Per-pixel Theil–Sen slope of growing-season annual values, 2015–2024, for (a) NDVI, (b) LST, (c) precipitation, (d) SRAD and (e) HFI. Each panel uses a diverging scale centred at zero and clipped at the 98th percentile of absolute values; the colour-bar label states what positive (+) and negative (–) values mean (NDVI and precipitation: green = increase, brown = decrease; LST and SRAD: red = increase, blue = decrease; HFI: orange = increase, purple = decrease). Significant Mann–Kendall trends ($p < 0.05$): NDVI 6.4%, LST 3.0%, precipitation 0.0%, SRAD 0.0%, HFI 39.9% of pixels.

The spatial patterns in Figure 3 were checked against the raster values by summarizing each slope layer in five 0.5° longitude bands between 65.1°E and 67.6°E . NDVI slope (Figure 3a) shows a clear west–east gradient: in the two western bands ($65.1\text{--}66.1^\circ\text{E}$; 5,666 valid pixels, 3–15% cropland) the median slope is -0.0048 and -0.0034 NDVI yr^{-1} and only 1–10% of pixels are greening, whereas in the two eastern bands ($66.6\text{--}67.6^\circ\text{E}$; 58–68% cropland) the median slope is $+0.0027$ and $+0.0032$ NDVI yr^{-1} and 78–81% of pixels are greening. Almost all individually significant NDVI trends are positive (5.9% of pixels significantly positive, 0.5% significantly negative). LST slope (Figure 3b) is negative almost everywhere (median -0.16 $^\circ\text{C yr}^{-1}$), with the strongest cooling in the far west (median -0.29 $^\circ\text{C yr}^{-1}$). Precipitation slopes (Figure 3c) are weakly positive nearly everywhere and largest in the centre, and SRAD slopes (Figure 3d) are small and positive, with a smooth south-eastward increase inherited from the coarse ERA5-Land grid; neither is significant at any pixel. HFI slope (Figure 3e) is strongly positive in the cropland-dominated centre and east (median $+0.42$ to $+0.53$ index yr^{-1} east of 66.6°E , where

40.5% of pixels have a significant increase) and near zero or slightly negative in the western rangelands (median -0.01 to -0.02 index yr^{-1} ; 31.8% of western pixels have a small but significant decrease). As shown in Section 4.7, most of the eastern HFI increase comes from the single 2018→2019 step.

4.2 RESTREND Model Fit and Residual Trend

The per-pixel climate regression ($\text{NDVI} \sim \text{LST} + \text{Precip} + \text{SRAD}$) explained a substantial fraction of interannual NDVI variance (Figure 4a).

Table 7. RESTREND goodness-of-fit across valid pixels.

Metric	Value
Mean R^2	0.664
Median R^2	0.745
Mean adjusted R^2	0.495
Median adjusted R^2	0.618
Fraction with $R^2 > 0.5$	75.9%
Pixels with adjusted $R^2 < 0$	2,671 (13.3%)

The median coefficients were -0.012 NDVI K^{-1} for LST, $+0.0011$ NDVI per mm month^{-1} for precipitation and -0.00025 NDVI per W m^{-2} for SRAD, with a median intercept of $+0.735$. The fit has a strong west–east structure (Figure 4a): the median R^2 is 0.16 and 0.43 in the two grassland-dominated western longitude bands (65.1 – 66.1°E) but 0.81–0.85 in the cropland-dominated centre and east (66.1 – 67.6°E). The 13.3% of pixels with negative adjusted R^2 are pixels with R^2 below $k/(n - 1) = 0.33$, where the three climate predictors explain no more variance than would be expected from three unrelated predictors.

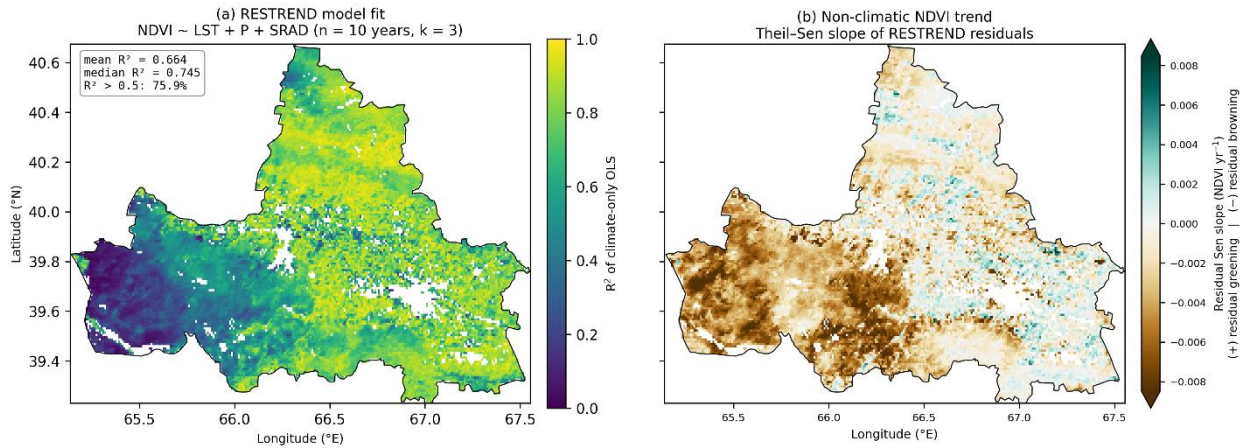


Figure 4. RESTREND results. (a) Per-pixel coefficient of determination (R^2) of the climate-only regression $NDVI \sim LST + P + SRAD$ ($n = 10$, $k = 3$). (b) Theil–Sen slope of the per-pixel residual time series, interpreted as the non-climatic component of NDVI change (green = residual greening, brown = residual browning; scale clipped at the 98th percentile). Mann–Kendall significance ($p < 0.05$) is reached at 10.1% of pixels.

The residual Sen slope (Figure 4b) isolates the non-climatic NDVI trend. Statistically significant residual trends occur at 10.1% of pixels (2,030 pixels), and the median residual slope is negative (-0.0019 NDVI yr^{-1}). Negative residual slopes dominate the western lowlands: west of $66.1^\circ E$ the median residual slope is -0.0051 and -0.0038 NDVI yr^{-1} in the two bands, and only 0–2% of pixels have a positive residual trend. Towards the east the residual trend approaches zero (median -0.0001 NDVI yr^{-1} at 66.6 – $67.1^\circ E$ and $+0.0004$ NDVI yr^{-1} at 67.1 – $67.6^\circ E$, with 48–62% of pixels positive), with scattered positive values in the irrigated cropland of the centre and east. The residual decline is therefore concentrated precisely where the climate model fits worst (Figure 4a).

4.3 Climate Contribution and Human Attribution

Per-pixel Climate Contribution (CC) and Human Attribution (HA) were computed on a shared symmetric scale.

CC (Figure 5a) is positive over most of the province (mean $+0.0022$, median $+0.0016$ NDVI yr^{-1} ; 86% of pixels positive east of $66.6^\circ E$), indicating that the climate trends of the decade — mainly weak LST cooling combined with a slight increase in precipitation — favoured vegetation. CC is slightly negative only in parts of the far west (median -0.0004 NDVI yr^{-1} at 65.1 – $65.6^\circ E$). HA (Figure 5b) is small in most of the centre and east (median $+0.0001$ to $+0.0003$ NDVI yr^{-1}) and negative in the western lowlands (median -0.0015 NDVI yr^{-1} west of $66.1^\circ E$, with 90–98%

of pixels negative). The negative western HA does not arise from rising human pressure: the HFI trend there is near zero or slightly negative (Section 4.1), and HA is negative because a slightly declining HFI is multiplied by a positive residual–HFI coefficient e . The median e is +0.0015 NDVI per HFI unit (mean +0.027, right-skewed), but it is much larger in the far west (median +0.135 at 65.1–65.6°E), where HFI hardly varies over time and e is therefore poorly constrained (Section 4.7). Across all pixels, the per-pixel median relative contributions are CC% = 82% and HA% = 18% (means 71% and 29%).

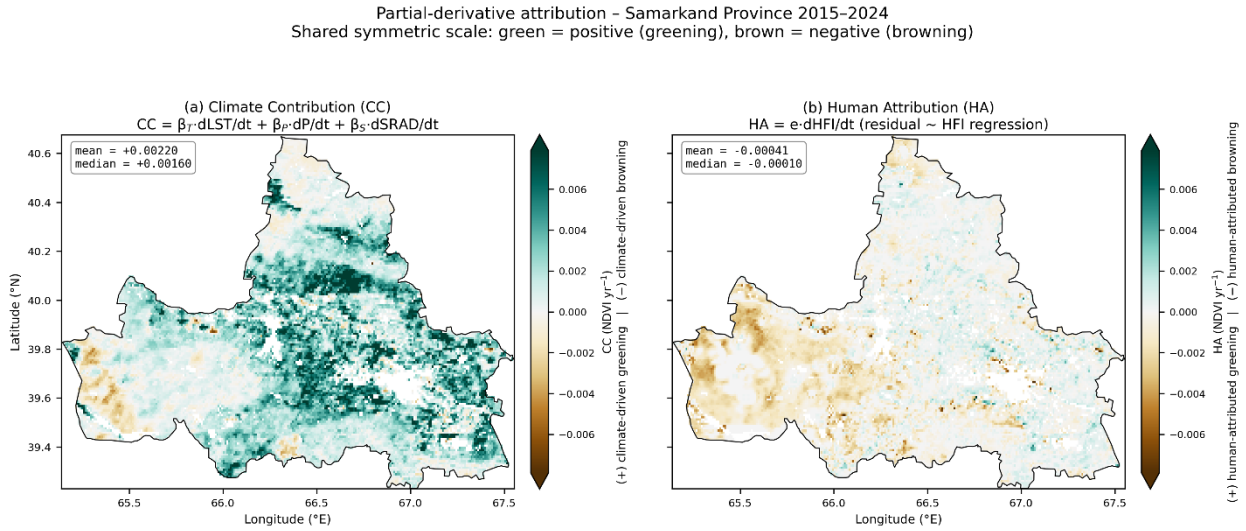


Figure 5. Partial-derivative attribution of the observed NDVI trend. (a) Climate Contribution (CC). (b) Human Attribution (HA). Both fields are in NDVI yr⁻¹ on a shared symmetric scale centred at zero (green = positive, i.e. contribution to greening; brown = negative, i.e. contribution to browning; clipped at the 98th percentile).

4.4 Six-Class Attribution Map

The headline six-class attribution map (Figure 6) classifies each pixel by the sign of its NDVI trend and the relative magnitudes of CC and HA (Table 8).

Table 8. Attribution class areas (pixel area $\approx 0.762 \text{ km}^2$ at the grid-centre latitude; $n \text{ valid} = 20,132$).

Class	Description	Pixels	% of valid	Area (km ²)
1	Climate-dominated restoration	5,268	26.2%	4,015
2	Human-dominated restoration	115	0.6%	88
3	Co-driven restoration	105	0.5%	80
4	Climate-dominated degradation	1,259	6.3%	960
5	Human-dominated degradation	4,030	20.0%	3,072
6	Co-driven degradation	352	1.7%	268
0	Unclassified	9,003	44.7%	6,862
Total valid		20,132	100%	15,345

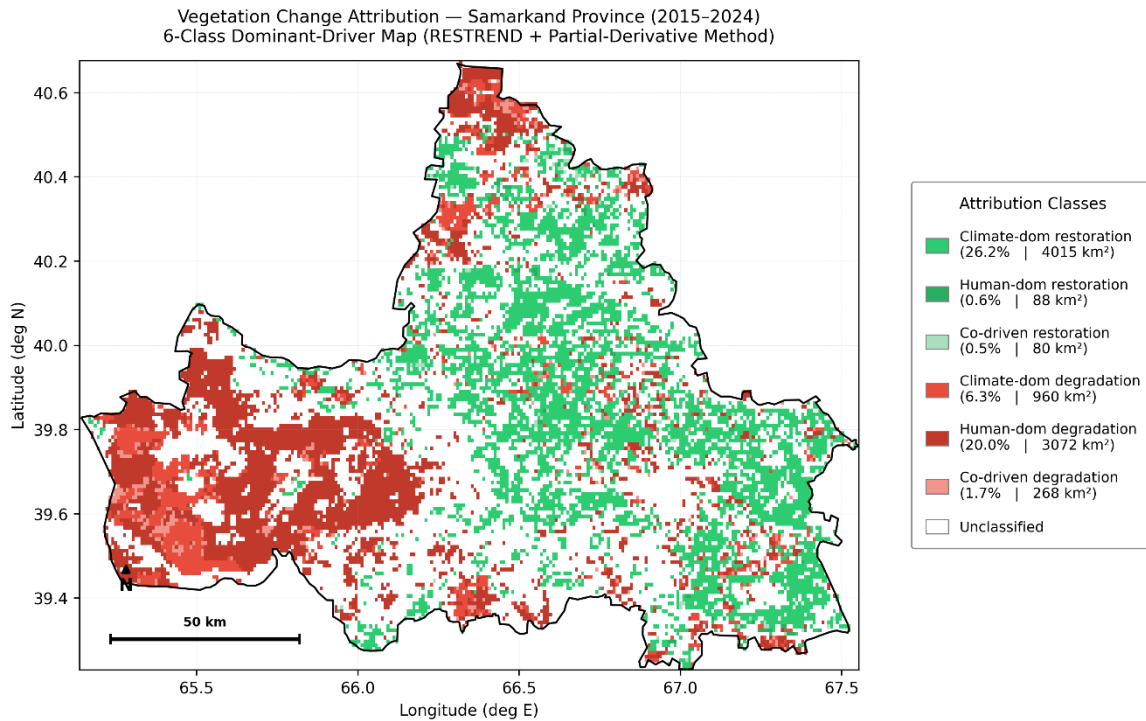


Figure 6. Six-class attribution map of vegetation change in Samarkand Province, 2015–2024. Green shades denote restoration classes (1–3) and red shades degradation classes (4–6); white denotes unclassified valid pixels as well as pixels outside the vegetation mask.

Among the 11,129 classified pixels (55.3% of valid), restoration classes (1–3) account for 5,488 pixels (49.3% of classified) and degradation classes (4–6) for 5,641 pixels (50.7%). The spatial arrangement, verified from the class raster, matches Figure 6. Class 5 (human-dominated degradation, 20.0% of valid pixels) is concentrated in the western lowlands: 2,594 of its 4,030 pixels (64%) lie west of 66.1°E, where they make up 46% of all valid pixels, and 87% of Class-5 pixels are grassland. Class 1 (climate-dominated restoration, 26.2%) occupies the central and eastern districts: 5,058 of its 5,268 pixels (96%) lie east of 66.1°E, and 76% of them

are cropland. Climate-dominated degradation (Class 4, 6.3%) also lies mainly in the west (mean longitude 66.06°E). The unclassified fraction (44.7%) comprises pixels in which the signs of the NDVI trend, CC and HA do not match any of the six rules — typically CC and HA of opposite sign.

4.5 Random Forest Feature Importances

The global Random Forest regressor, trained on the pooled pixel–year dataset with a spatial 80/20 hold-out split, achieved strong predictive performance on the held-out test set (Table 9).

Table 9. Random Forest test-set performance.

Metric	Value
Test R ²	0.925
Test RMSE	0.0335 NDVI
Test MAE	0.0238 NDVI

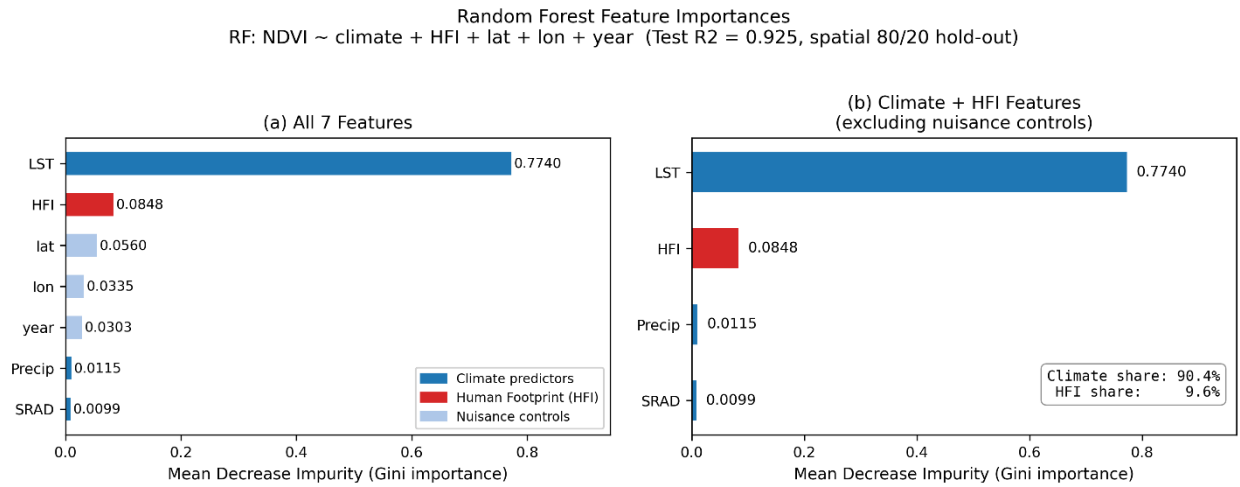


Figure 7. Random Forest feature importance (Mean Decrease Impurity) for the pooled pixel–year prediction of NDVI

Gini feature importances ranked LST first by a wide margin (0.774), followed by HFI (0.085), the spatial and temporal nuisance controls (lat 0.056, lon 0.034, year 0.030), and finally precipitation (0.012) and SRAD (0.010). Among the four physical drivers only, the three climate variables collectively account for 90.4% of importance and HFI for 9.6%. This is broadly consistent with the RESTREND partial-derivative attribution (median CC $\approx$ 82%, HA $\approx$ 18%); both methods rank

climate — and specifically LST — as the dominant driver, while differing modestly in the human share.

4.6 Independent Validation: Residual Slope vs. HFI Slope

The RESTREND residual Sen slope (non-climatic NDVI trend) was correlated against the HFI Sen slope across all 20,132 valid pixels.

Table 10. *Residual–HFI validation statistics.*

Statistic	Value
n	20,132
Pearson r	+0.338 ($p < 10^{-16}$)
Spearman ρ	+0.383 ($p < 10^{-16}$)
OLS slope	+34.8
OLS R ²	0.114

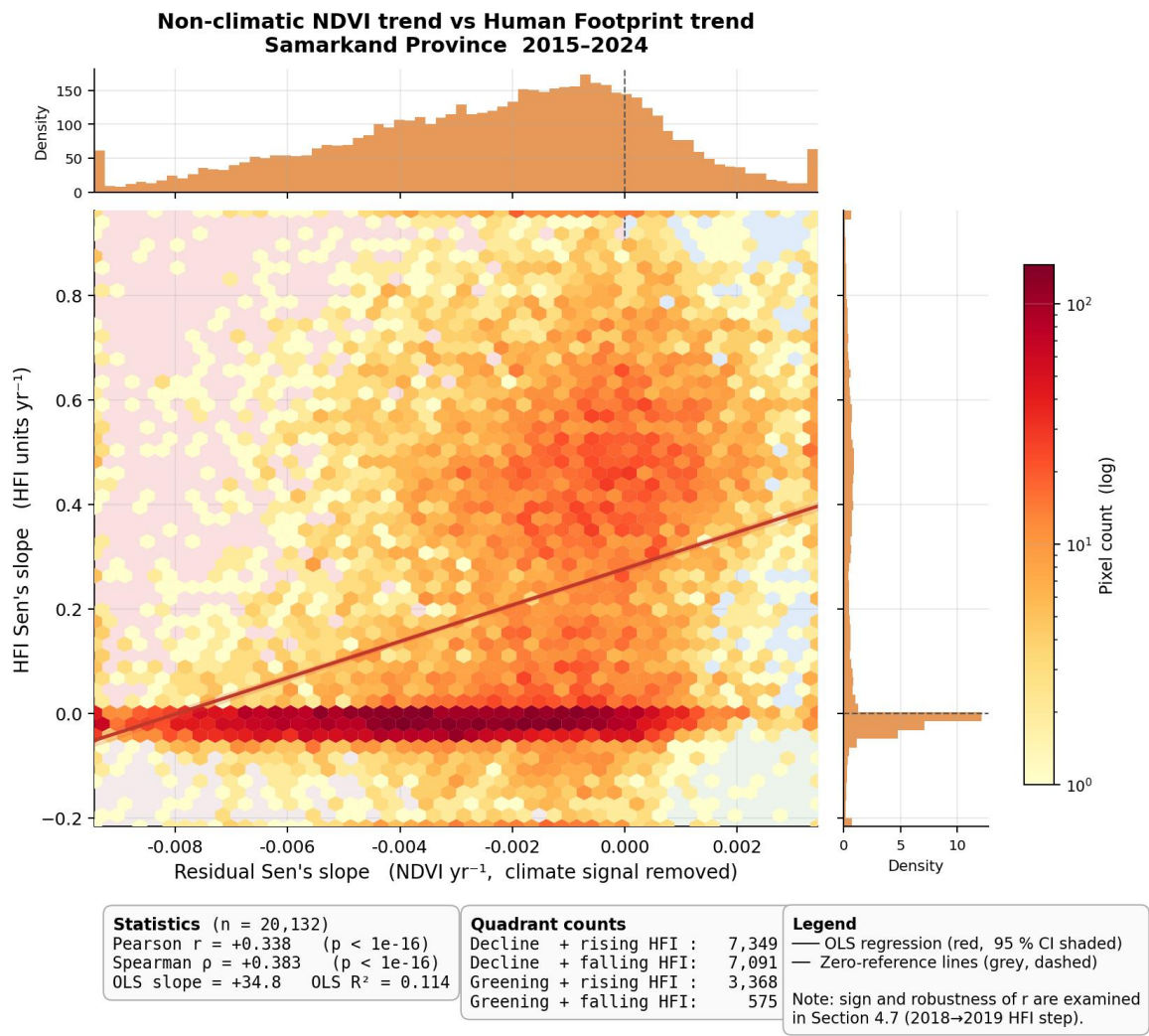


Figure 8. Hexbin scatter of per-pixel RESTREND residual Sen slope (x-axis) against HFI Sen slope (y-axis).

The correlation is statistically significant but positive — the opposite sign to the hypothesized negative relationship (more HFI growth → more vegetation loss). Most pixels lie in a dense band near HFI slope ≈ 0 across the full range of residual slopes, while a right-skewed minority of pixels with strongly rising HFI — located in the cropland-dominated centre and east — have residual slopes close to zero; this asymmetry produces the positive correlation. The largest quadrant (36.5% of pixels) combines residual decline with rising HFI, and the second largest (35.2%) combines residual decline with falling HFI, which corresponds largely to the western rangelands. The OLS R^2 of 0.114 indicates that the HFI trend explains about 11% of the spatial variance of the residual trend. Because the high-HFI-growth pixels are exactly those affected by the 2018→2019 HFI step, the correlation is re-examined in Section 4.7.

4.7 Robustness to the 2018–2019 HFI Discontinuity

Figure 9 shows the discontinuity that motivates this analysis. HFI is nearly flat from 2015 to 2018 in all zones, jumps in 2019 in the centre and east, and is nearly flat again afterwards, whereas the western rangelands show almost no change. Table 11 summarizes how the HFI-dependent results respond to the two alternative treatments of the series described in Section 3.5.7.

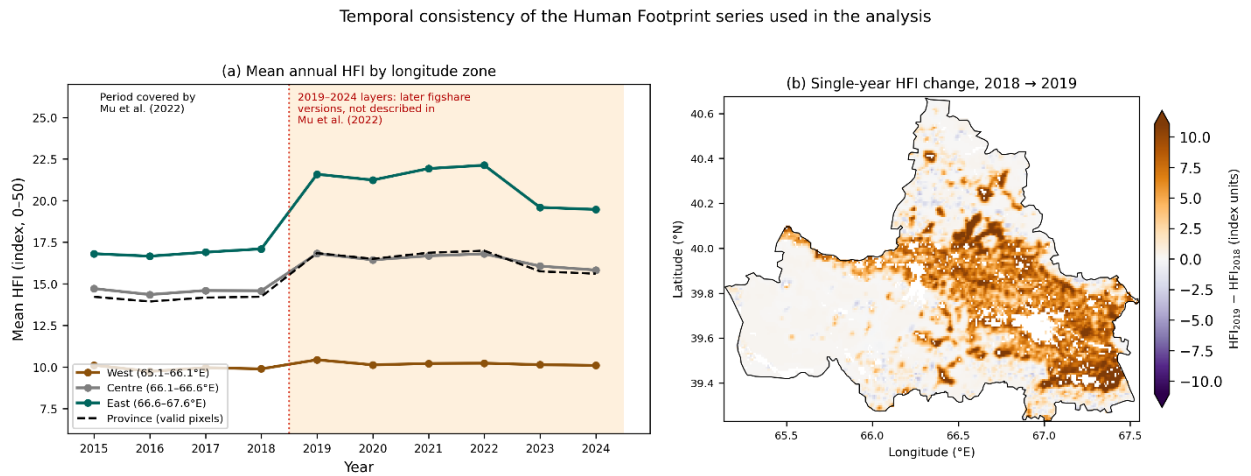


Figure 9. Temporal consistency of the HFI series used in the analysis. (a) Mean annual HFI of valid pixels in three longitude zones and for the province; the shaded period (2019–2024) is covered only by later versions of the figshare record, not by Mu et al. (2022). (b) Per-pixel HFI change between 2018 and 2019 (orange = increase, purple = decrease).

Table 11. Sensitivity of HFI-dependent results to the treatment of the 2018→2019 HFI discontinuity ($n = 20,132$ valid pixels). S0: original series; S1: step-adjusted; S2: frozen at 2018.

Quantity	S0 original	S1 step-adjusted	S2 frozen at 2018
Mean HFI Sen slope (index yr ⁻¹)	+0.198	−0.020	+0.004
Pixels with significant HFI trend, increase / decrease (%)	22.6 / 17.3	3.5 / 42.2	9.6 / 1.8
Median HA share, HA% (%)	18.0	18.5	0.0
Class 1, climate-dominated restoration (% of valid)	26.2	15.4	3.0
Class 4, climate-dominated degradation (% of valid)	6.3	4.0	14.9
Class 5, human-dominated degradation (% of valid)	20.0	27.3	1.4
Class-5 pixels west of 66.1°E	2,594	3,752	77
Unclassified (% of valid)	44.7	51.5	80.5
Residual vs HFI slope: Pearson r / Spearman ρ	+0.338 / +0.383	+0.062 / +0.173	+0.102 / +0.180
RF HFI importance (share of physical drivers)	0.085 (9.6%)	0.067 (7.7%)	0.079 (9.1%)
RF test R ²	0.925	0.924	0.924

Three results are robust. First, the dominance of climate does not depend on the HFI treatment: the median HA share stays at 18–19% whenever HFI is allowed to vary (S0, S1), and the Random Forest ranks HFI second with 7.7–9.6% of the physical-driver importance and an unchanged test R² in all scenarios. Second, the west–east contrast in NDVI trend, residual trend, model fit and CC does not involve HFI at all and is therefore unaffected. Third, the western concentration of Class 5 persists and even strengthens under S1 (3,752 western Class-5 pixels), because the step is almost absent in the west.

Two results are not robust. (i) The positive residual–HFI correlation largely disappears once the step is removed (Pearson r falls from +0.338 to +0.062 and Spearman ρ from +0.383 to +0.173). Most of the original correlation therefore reflects the spatial coincidence between the eastern croplands, where the 2019 HFI step is large and the residual trend is near zero, and the western rangelands, where the step is absent and the residual trend is negative — not a demonstrated link between human-footprint growth and residual greening. (ii) The areal split between classes depends strongly on the HFI series: Class 1 shrinks from 26.2% to 15.4% under S1, because removing the step makes the eastern HFI trends slightly negative

and hence HA negative, and under S2 most pixels become unclassified because HA collapses to zero.

A further diagnostic concerns the reliability of the Class-5 label. In Class-5 pixels the median temporal standard deviation of HFI over the ten years is only 0.16 index units (2.73 in Class 1), and the residual–HFI regression is significant at $p < 0.05$ in only 14.2% of them. The large negative HA values in the west therefore come from a very small, slightly negative HFI trend multiplied by a poorly constrained coefficient e . What the data robustly show in the west is a non-climatic NDVI decline: a negative residual trend where the climate model explains little of the interannual variance. The decision rule labels this decline “human-dominated” because HA is the only non-climatic term in the decomposition, but the HFI data cannot confirm that human pressure is its cause. Class 5 is therefore best read as non-climatic (human-attributed) degradation, and its causes — for example grazing pressure that HFI does not resolve, reduced irrigation return flow, groundwater decline, or lagged drought effects outside the growing-season climate window — require independent verification.

5. Discussion and Conclusion (Critical Thinking Section)

5.1 Key Findings

Climate is the dominant driver of vegetation variability, with human pressure a secondary factor. Both analytical approaches converge on this hierarchy. The RESTREND partial-derivative attribution gives a per-pixel median climate share of approximately 82% (CC%) against 18% for the human term (HA%), while the independent Random Forest ranks Land Surface Temperature as the single most important predictor by a wide margin (Gini importance 0.774) and assigns the three climate variables 90.4% of the importance among the four physical drivers, against 9.6% for HFI (Figure 7). Both results are stable under the alternative HFI treatments of Section 4.7 (HA% 18–19%; HFI importance 7.7–9.6%). The agreement of two methodologically distinct models on the same hierarchy is a strong internal-consistency result. A non-climatic signal is nonetheless substantial: human-dominated degradation (Class 5) is the second-largest class, covering 20.0% of valid pixels (~3,072 km²), although its attribution specifically to human pressure is uncertain (see below).

A pronounced west–east gradient structures the province. Read from zonal statistics of the rasters, every layer shows the same pattern. The grassland-dominated western lowlands (west of ~66.1°E; 3–15% cropland) combine declining NDVI

(median Sen slope -0.0034 to -0.0048 NDVI yr^{-1} ; Figure 3a), strongly negative residual trends (median -0.0038 to -0.0051 NDVI yr^{-1} ; Figure 4b), poor climate-model fit (median R^2 0.16–0.43; Figure 4a), negative HA (Figure 5b) and most of the Class-5 pixels (dark red in Figure 6). The cropland-dominated central and eastern districts (66.1 – 67.6°E ; 46–68% cropland) combine greening (median slope up to $+0.0032$ NDVI yr^{-1}), near-zero residual trends, high R^2 (0.81–0.85), positive CC and a dominance of Class 1 (green in Figure 6), consistent with the weak regional LST cooling and slight precipitation increase acting on irrigated, climate-coupled cropland. All directional statements in this report were derived from zonal statistics in 0.5° longitude bands rather than from visual inspection of the maps, and every map carries an explicit colour-bar legend.

The climate-regression model performs well where vegetation is climate-coupled, and poorly in the western rangelands. The RESTREND R^2 map (Figure 4a) shows high explanatory power (median R^2 0.81–0.85) across the central and eastern province but conspicuously low R^2 (median 0.16) in the far western lowlands — the region where the attribution map places most human-dominated degradation. Where a climate-only model fails to explain NDVI, a non-climatic driver is implicated. That driver may be human land use, but it may also be a climatic or hydrological control that the three growing-season predictors do not capture, such as winter snowpack, soil-moisture memory or groundwater. The low R^2 therefore supports the existence of a non-climatic signal in the west, not its human origin.

The RF and RESTREND human-share estimates differ in magnitude, and the difference is methodologically expected. RF assigns HFI 9.6% importance; RESTREND assigns HA 18% of the trend attribution. The gap arises because the two methods answer different questions. RF Gini importance reflects the variable's contribution to explaining all NDVI variance — overwhelmingly the large cross-sectional spatial gradient between high-LST dryland and low-LST irrigated land, which LST captures almost entirely. RESTREND, by contrast, first removes the climate signal and then attributes only the residual temporal trend, in which the human signal is proportionally larger. The two numbers are therefore not contradictory; they bracket the human contribution from two directions (≈ 10 –18%).

The residual-versus-HFI validation does not provide independent support for the human attribution. The original analysis yielded a statistically significant but positive correlation (Pearson $r = +0.338$, Spearman $\rho = +0.383$, OLS $R^2 = 0.114$), opposite in sign to the naïve degradation hypothesis, which could be read as irrigation-driven agricultural intensification raising NDVI where HFI grows. The sensitivity analysis (Section 4.7) shows that the data do not support this reading as they stand: once the 2018→2019 step in the HFI record is removed, the

correlation falls to $r = +0.062$ ($\rho = +0.173$). The positive correlation mainly reflects the spatial coincidence between the eastern croplands, where the HFI step is large and the residual trend is near zero, and the western rangelands, where the step is absent and the residual trend is negative. Irrigation expansion remains a plausible hypothesis for the east, since the climate model contains no irrigation term, but testing it requires temporally consistent human-pressure data — ideally component-level cropland, irrigation and grazing layers — rather than the composite HFI.

5.2 Limitations

Weak Mann–Kendall power at $n = 10$. Only 6.4% of NDVI pixels reach individual significance at $p < 0.05$, because the test requires $|\tau| > 0.51$ for a ten-point series. Sen's slopes are therefore presented as best estimates, and a non-significant pixel should not be read as having no trend. This is the standard limitation of a decadal RESTREND analysis.

Annual temporal resolution. Following standard RESTREND protocol, the analysis uses growing-season annual means and cannot resolve phenological shifts, irrigation scheduling, or sub-annual lag effects. These would require a monthly or seasonal time series [1][2].

MODIS LST is surface, not air, temperature. Daytime LST at the ~10:30 overpass can exceed 2-metre air temperature by 10–20 °C on hot days. As a proxy for plant physiological stress, air temperature would be biologically more appropriate, but LST is the only spatially continuous, directly measured thermal field available at the required resolution. Because LST is also the dominant predictor in both models, this caveat propagates into the attribution and should be borne in mind [3][4].

No per-pixel quality screening of the MODIS inputs. The MOD13Q1 and MOD11A2 composites were used without additional filtering by their quality layers (SummaryQA, DetailedQA and pixel reliability for NDVI; QC_Day for LST). The analysis therefore relies on the quality selection of the compositing algorithms and on the April–October averaging. Residual cloud or snow contamination in individual composites, particularly in April and October in the uplands, may add noise to the annual values, and assigning composites to months by their start date slightly blurs monthly boundaries. Repeating the analysis with QA-filtered composites is a priority for a future version.

CHIRPS winter precipitation bias. CHIRPS underestimates solid winter precipitation because the cold-cloud-duration algorithm is insensitive to snowfall.

The April–October aggregation used here largely avoids this, but early-season soil moisture from spring snowmelt may be under-represented.

ERA5-Land SRAD is reanalysis model output. SSRD is not a direct satellite measurement and cannot resolve terrain shading at its native ~11 km resolution. Its very low RF importance (0.0099) confirms it adds little explanatory power beyond LST and precipitation in this setting [16].

Temporal consistency and composition of HFI. The HFI layers for 2019–2024 come from later versions of the Mu et al. figshare record whose production method is not documented in the peer-reviewed paper, and the series contains a 2018→2019 step about five times larger than a typical annual change (Section 2.2.5). The sensitivity analysis (Section 4.7) shows that the class areas and the residual–HFI correlation depend on how this break is treated, while the climate-versus-human hierarchy does not. In addition, the composite uses globally fixed layer weights that may not suit Samarkand's agriculture–pastoralism–urban mix, several component layers appear not to be updated every year, and the index aggregates greening-associated pressures (irrigation, cropland) and degradation-associated pressures (grazing, built-up area) into a single number. Where HFI barely changes, the residual–HFI coefficient is poorly constrained, which makes the Class-5 label uncertain [10][11][19].

OLS linearity assumption. RESTREND assumes a linear, time-invariant NDVI–climate relationship. Non-linear responses (e.g. NDVI saturation, threshold drought effects) and changing sensitivities are not captured. The Random Forest cross-check partially mitigates this by fitting a non-parametric model, but the attribution itself still rests on the linear coefficients.

Spatial autocorrelation. Pixels within the same irrigation scheme or landscape unit share drivers and responses, so treating each pixel-year as independent overstates the effective sample size. The reported p-values are therefore nominal; significance accounting for spatial autocorrelation would be more conservative.

Correlation is not causation, and the unclassified fraction is large. Positive HFI–NDVI associations in irrigated areas cannot establish causation, since both respond to irrigation investment. Furthermore, 44.7% of valid pixels remain unclassified because CC and HA have opposing signs or near-zero magnitudes without a dominant driver; the six-class map therefore characterizes a little over half the province confidently, and the unclassified remainder should not be interpreted as "stable".

5.3 Potential Improvements

Several extensions would strengthen a future iteration of this work. The most urgent is a temporally consistent human-pressure series: repeating the analysis for 2000–2018 with the documented Mu et al. release and the full MODIS archive would remove the 2019 discontinuity and almost double the record length, and replacing the composite HFI with harmonized component layers — gridded population [30], cropland and irrigation extent, and grazing pressure — would allow a fair test of the degradation hypothesis that the aggregate index currently obscures. The MODIS inputs should be quality-screened with their QA layers before compositing. A structural-break method such as BFAST applied before trend estimation would separate gradual change from abrupt events, notably the drought years 2018 and 2021 evident in the interannual NDVI record, and would flag breaks in driver series such as the 2019 HFI step. Adding GRACE/GRACE-FO groundwater anomalies and winter precipitation or snow-cover predictors would address unmodelled hydrological controls, which are plausible alternative explanations for the non-climatic decline in the west. Replacing the per-pixel OLS with geographically weighted regression, or with causal-inference methods designed for short, autocorrelated time series [27][28], would relax the pixel-independence and linearity assumptions. Finally, substituting a directly satellite-derived radiation product (e.g. MODIS MCD18A1) for the ERA5-Land reanalysis SRAD would remove the one model-dependent input in the current stack.

Conclusion

This study applied an integrated RESTREND and partial-derivative attribution framework, cross-checked with a Random Forest model, a residual–HFI comparison and a sensitivity analysis of the Human Footprint record, to quantify the roles of climate variability and anthropogenic pressure in the vegetation dynamics of Samarkand Province over 2015–2024. Four conclusions emerge. First, climate variability — specifically Land Surface Temperature — is the dominant control on NDVI: the per-pixel median climate share of the attributable trend is about 82% in the RESTREND framework, climate variables carry about 90% of the physical-driver importance in the Random Forest, and both figures are insensitive to how the HFI discontinuity is treated. Second, the province is organized along a robust west–east gradient: the grassland-dominated western lowlands show an NDVI decline that the climate model cannot explain (negative residual trends, median R^2 as low as 0.16), whereas the cropland-dominated central and eastern districts show climate-attributable greening. Third, the decision rule labels the western decline human-dominated degradation (20.0% of valid pixels), but HFI barely changes there and the residual–HFI relationship is weak, so this label identifies a non-climatic decline whose human cause remains to be confirmed. Fourth, the HFI record used for 2019–2024 comes from an undocumented extension of the Mu et al. dataset and contains a strong 2018→2019 step; the apparent positive link between HFI growth and residual greening largely disappears when this step is removed, which shows both the value of sensitivity testing and the limits of a composite, temporally inconsistent human-pressure index for attribution. Together these results provide a spatially explicit evidence base for land and water management in Samarkand Province, while clearly separating the conclusions that are robust from those that require better human-pressure data.

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List of Abbreviations

Abbreviation	Definition
AOI	Area of Interest
BFAST	Breaks For Additive Season and Trend
BRDF	Bidirectional Reflectance Distribution Function
C6.1	MODIS Collection 6.1
CCD	Cold-Cloud Duration
CC	Climate Contribution (RESTREND partial-derivative attribution)
CHIRPS	Climate Hazards group InfraRed Precipitation with Stations
CRS	Coordinate Reference System
GDAL	Geospatial Data Abstraction Library
GEE	Google Earth Engine
HA	Human Attribution (RESTREND partial-derivative attribution)
HFI	Human Footprint Index
IFS	Integrated Forecasting System (ECMWF)
LST	Land Surface Temperature
MAE	Mean Absolute Error
MDI	Mean Decrease Impurity (Random Forest feature importance)
MK	Mann-Kendall (trend test)

MODIS	Moderate Resolution Imaging Spectroradiometer
NDVI	Normalised Difference Vegetation Index
NIR	Near-Infrared (reflectance band, ~850 nm)
OLS	Ordinary Least Squares (regression)
PAR	Photosynthetically Active Radiation
QA	Quality Assessment (MODIS quality layers)
R ²	Coefficient of Determination
RESTREND	Residual Trend Analysis
RF	Random Forest
RMSE	Root Mean Square Error
SAR	Synthetic Aperture Radar
Sen	Theil-Sen slope estimator
SRAD	Surface Solar Radiation Downwelling
SSRD	Surface Solar Radiation Downwelling (ERA5 variable name)
WGS84	World Geodetic System 1984 (datum, EPSG:4326)